07/16/2026

Quiz 3

NAME: ______________________________

Q.1. Express $4x^2+8x+5$ in completed square form (i.e. as $a(x+b)^2+c$ where a, b, c are constants).

We start by factoring out the coefficient of x^2 from the two x terms

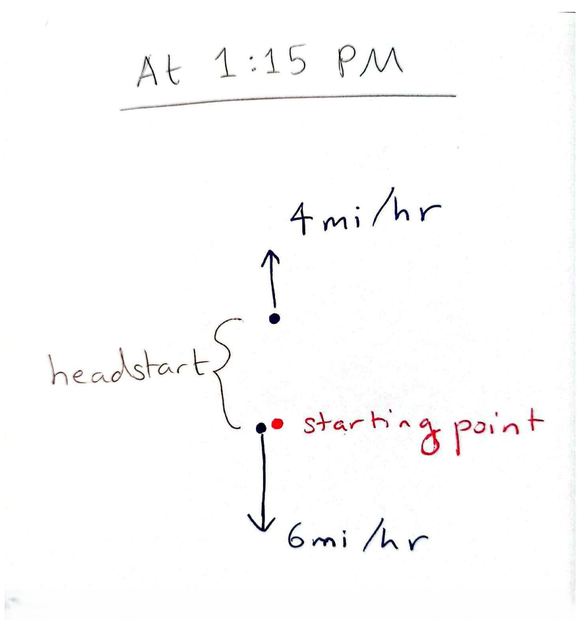
$$4x^2 + 8x + 5 = 4(x^2 + 2x) + 5$$

Now inside the bracket we have $x^2 + 2x$. If we add 1 to this, we get $x^2 + 2x + 1 = (x + 1)^2$, which is our desired completed square form, so:

$$\begin{aligned} 4x^2 + 8x + 5 &= 4(x^2 + 2x) + 5 \\ &= 4(\underbrace{x^2 + 2x + 1}_{(x+1)^2} - 1) + 5 \\ &= 4((x + 1)^2 - 1) + 5 \\ &= \boxed{4(x + 1)^2 + 1} \end{aligned}$$

Q.2. Two children own two-way radios that have a maximum range of 2 miles. One leaves a certain point at 1 : 00 P.M., walking due north at a rate of 4 mi/hr. The other leaves the same point at 1 : 15 P.M., traveling due south at 6 mi/hr. When will they be unable to communicate with one another?

It helps to draw a picture:



Since the child going north leaves 15 mins earlier than the child going south, they have a headstart. Let's try and calculate this headstart. 15 mins = 0.25 hr and the child going north has a speed of 4 mi/hr, so in that 15 mins they were able to cover an extra (4 mi/hr) × (0.25 hr) = 1 mi.

So the child going north is already 1 mi north of the starting point when the child going south starts walking. Now, let t be the time in hours after $1:15$. The distance in miles between the two children at time t is:

$$(4t + 1) + 6t = 10t + 1$$

The moment this distance equals 2, the radios will hit their maximum range and the children will not be able to communicate with each other. So we need to solve $10t + 1 = 2$. This gives us $t = 0.1$ hr. Remember that t is the time in hours after $1:15$. In minutes, t is 6 mins, so the time at which the children will be unable to communicate is $\boxed{1:21 \text{ P.M.}}$.

Q.3. Express the complex fraction $\frac{3+5i}{1+i}$ as a single complex number (i.e. as $a + bi$ where a and b are real).

We just need to multiply and divide by the conjugate of the denominator (remember that conjugate of $a + bi$ is $a - bi$ and that the product of a complex number with its conjugate is its 'length' squared, i.e. $(a + bi)(a - bi) = a^2 + b^2$).

The denominator is $1 + i$. This has conjugate $1 - i$, so:

$$\begin{aligned}
\frac{3+5i}{1+i} &= \frac{3+5i}{1+i} \cdot \frac{1-i}{1-i} \\
&= \frac{(3+5i)\cdot(1-i)}{(1+i)\cdot(1-i)} \\
&= \frac{3\cdot 1 + 3\cdot(-i) + 5i\cdot 1 + 5i\cdot(-i)}{1^2 + 1^2} \\
&= \frac{3 - 3i + 5i + 5}{1+1} \\
&= \frac{8+2i}{2} = \boxed{4+i}
\end{aligned}$$

Q.4. Find all real solutions to the equation $2|5x + 2| - 1 = 5$.

Rearranging, we get:

$$\begin{aligned}
&2|5x + 2| - 1 = 5 \\
\Rightarrow &2|5x + 2| = 6 \\
\Rightarrow &|5x + 2| = 3
\end{aligned}$$

Now recall that $|y| = a$ for any $a \geq 0$, means $y = \pm a$. In this case, we have that:

$$
\begin{aligned}
5x + 2 &= 3 && \text{OR} && 5x + 2 = -3 \\
\Rightarrow 5x &= 1 && \text{OR} && 5x = -5 \\
\Rightarrow x &= \frac{1}{5} && \text{OR} && x = -1
\end{aligned}
$$

So $\boxed{x = \frac{1}{5} \text{ OR } x = -1}$.